resolved into partial fractions by taking the numarator of ax^2+bx+c of the form Ax+B.

Example 2.7

Resolve into partial fractions:

$$\frac{2x+1}{(x-1)(x^2+1)}$$

Solution

Here x^2+1 cannot be factorized into linear factors.

Let $\frac{2x+1}{(x-1)(x^2+1)} = \frac{A}{x-1}+\frac{Bx+c}{x^2+1}$ ... (1)

Multiplying both sides by $(x-1)(x^2+1)$

$$2x+1 = A(x^2+1)+(Bx+C)(x-1) \text{ ... (2)}$$

Put $x = 1$ in (2)

$$2+1 = A(1+1)+0$$

$$\therefore A = \frac{3}{2}$$

Put $x = 0$ in (2)

$$0+1 = A(0+1)+(0+C)(-1)$$

$$1 = A - C$$

$$\therefore C = \frac{1}{2}$$

Equating the coefficient of x^2 on both the sides of (2), we get

$$A+B = 0$$

$$B = -A$$

$$\therefore B = \frac{-3}{2}$$

Substituting the values of A, B and C in (1), we get

$$\frac{2x+1}{(x-1)(x^2+1)} = \frac{\frac{3}{2}}{x-1}+\frac{\frac{-3}{2}x+\frac{1}{2}}{x^2+1}$$

$$= \frac{3}{2(x-1)}-\frac{3x-1}{2(x^2+1)}$$

Exercise 2.1

Resolve into partial fractions for the following:

1. $\frac{3x+7}{x^2-3x+2}$
2. $\frac{4x+1}{(x-2)(x+1)}$
3. $\frac{1}{(x-1)(x+2)^2}$
4. $\frac{1}{x^2-1}$
5. $\frac{x-2}{(x+2)(x-1)^2}$
6. $\frac{2x^2-5x-7}{(x-2)^3}$
7. $\frac{x^2-6x+2}{x^2(x+2)}$
8. $\frac{x^2-3}{(x+2)(x^2+1)}$
9. $\frac{x+2}{(x-1)(x+3)^2}$
10. $\frac{1}{(x^2+4)(x+1)}$

2.2 Permutations

2.2.1 Factorial

For any natural number n, 'n factorial' is defined as the product of the first n natural numbers and is denoted by $n!$ or $\underline{|n}$.

For any natural number n,
$n! = n(n-1)(n-2) \ldots 3\times2\times1$

For examples,

$5! = 5\times4\times3\times2\times1 = 120$

$4! = 4\times3\times2\times1 = 24$

$3! = 3\times2\times1 = 6$

$2! = 2\times1 = 2$

$1! = 1$

NOTE

$0! = 1$

For any natural number n

$$n! = n(n-1)(n-2) \ldots 3\times2\times1$$

$$= n(n-1)!$$

$$= n(n-1)(n-2)!$$

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